

ORIGINAL ARTICLE

Ketamine or Etomidate for Tracheal Intubation of Critically Ill Adults

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ABSTRACT

BACKGROUND

For critically ill adults undergoing tracheal intubation, observational studies suggest that the use of etomidate to induce anesthesia may increase the risk of death. Whether the use of ketamine rather than etomidate decreases the risk of death is uncertain.

METHODS

In a randomized trial conducted in 14 emergency departments and intensive care units in the United States, we randomly assigned critically ill adults who were undergoing tracheal intubation to receive ketamine or etomidate for the induction of anesthesia. The primary outcome was in-hospital death from any cause by day 28. The secondary outcome was cardiovascular collapse during intubation, defined by the occurrence of a systolic blood pressure below 65 mm Hg, receipt of a new or increased dose of vasopressors, or cardiac arrest.

RESULTS

A total of 2365 patients underwent randomization and were included in the trial population; 1176 were assigned to the ketamine group and 1189 to the etomidate group. In-hospital death by day 28 occurred in 330 of 1173 patients (28.1%) in the ketamine group and in 345 of 1186 patients (29.1%) in the etomidate group (risk difference adjusted for trial site, −0.8 percentage points; 95% confidence interval [CI], −4.5 to 2.9; $P=0.65$). Cardiovascular collapse during intubation occurred in 260 of 1176 patients (22.1%) in the ketamine group and in 202 of 1189 patients (17.0%) in the etomidate group (risk difference, 5.1 percentage points; 95% CI, 1.9 to 8.3). Prespecified safety outcomes were similar in the two groups.

CONCLUSIONS

Among critically ill adults undergoing tracheal intubation, the use of ketamine to induce anesthesia did not result in a significantly lower incidence of in-hospital death by day 28 than etomidate. (Funded by the Patient-Centered Outcomes Research Institute and others; RSI ClinicalTrials.gov number, NCT05277896.)

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*A full list of the RSI investigators in this trial is provided in the Supplementary Appendix, available at NEJM.org.

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MORE THAN 13 MILLION CRITICALLY ill adults undergo emergency tracheal intubation worldwide each year,^{1,2} and approximately 30% of these patients die before hospital discharge.^{3,4} Nearly all patients undergoing tracheal intubation in an emergency department (ED) or intensive care unit (ICU) receive a medication to induce anesthesia for the procedure.^{3,4} Whether the choice of induction medication for critically ill adults affects the risk of death or other outcomes is uncertain.

Etomidate is the medication most often used to induce anesthesia during emergency tracheal intubation in the United States.⁴⁻¹⁰ This imidazole-derived sedative-hypnotic agent, which acts on γ -aminobutyric acid receptors, has been described as an ideal induction medication for critically ill adults because of its rapid onset and limited effect on blood pressure and heart rate.¹¹⁻¹³ However, etomidate inhibits 11 β -hydroxylase in the adrenal glands, and a single dose decreases cortisol production for up to 72 hours.¹⁴⁻¹⁹ Concern that etomidate-induced corticosteroid insufficiency may cause organ dysfunction and death,^{7,17} particularly in patients with sepsis,²⁰⁻²² has led regulators to remove etomidate from the market in multiple countries.²³⁻²⁵

Ketamine, a dissociative agent that acts on N-methyl-D-aspartate receptors and does not impair cortisol production, is an increasingly used alternative to etomidate for induction of anesthesia in critically ill patients.^{7,26,27} Because its administration increases plasma catecholamine concentrations, ketamine has been postulated to maintain hemodynamic stability during intubation better than other induction medications.^{24,28,29} However, ketamine is also a negative inotrope and vasodilator,³⁰⁻³² and observational studies have shown an association between the receipt of ketamine and hypotension,^{26,33-35} arrhythmia,^{36,37} and cardiac arrest during intubation.³⁸⁻⁴⁰

Previous small and moderate-sized randomized trials and meta-analyses in which ketamine was compared with etomidate have shown conflicting results, with some suggesting that the use of ketamine decreases the risk of death⁴¹⁻⁴³ and others showing no marked differences in outcomes.^{18,44,45} To determine the effects of the use of ketamine, as compared with etomidate, for induction of anesthesia during emergency tracheal intubation, we conducted the Randomized Trial

of Sedative Choice for Intubation (RSI). We hypothesized that the use of ketamine would decrease the incidence of death.

METHODS

TRIAL DESIGN AND OVERSIGHT

The Pragmatic Critical Care Research Group^{4,8,46} conducted this pragmatic, multicenter, unblinded, randomized, parallel-group trial in which the use of ketamine was compared with the use of etomidate for induction of anesthesia during emergency tracheal intubation of critically ill adults. The trial was initiated by the investigators and approved by the institutional review board at Vanderbilt University Medical Center and the Food and Drug Administration. The trial was conducted under the exception from informed-consent requirements for emergency research; details are provided in the Supplementary Appendix, available with the full text of this article at NEJM.org.⁴⁷ The trial was registered at ClinicalTrials.gov before initiation and was overseen by an independent data and safety monitoring board. The trial protocol and statistical analysis plan, available at NEJM.org, were published before the conclusion of enrollment.⁴⁸ The authors vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol. The funding institutions had no role in the conception, design, or conduct of the trial; the collection, management, analysis, interpretation, or presentation of the data; or the preparation, review, or approval of the manuscript.

TRIAL SITES AND PATIENT POPULATION

The trial was conducted at 14 sites (6 EDs and 8 ICUs) in 6 medical centers across the United States. Critically ill patients 18 years of age or older who were undergoing tracheal intubation with the use of a medication to induce anesthesia were eligible. Patients were excluded if they were known to be pregnant, were known to be prisoners, were presenting with a primary diagnosis of trauma, or had an immediate need for tracheal intubation that precluded randomization. Patients were also excluded if the treating clinicians determined that the use of ketamine or etomidate was either necessary or contraindicated. Details of the trial sites and a complete list of the inclusion and exclusion criteria are provided in the Supplementary Appendix.

RANDOMIZATION

Patients were randomly assigned in a 1:1 ratio to receive ketamine or etomidate for the induction of anesthesia during tracheal intubation. Randomization was performed with the use of permuted blocks of variable size and was stratified according to trial site. The trial-group assignments were placed in sequentially numbered, opaque envelopes and remained concealed until after enrollment. Clinicians and research personnel were aware of the trial-group assignments after randomization.

INTERVENTIONS

For patients assigned to the ketamine group, clinicians were instructed to administer ketamine intravenously to induce anesthesia for tracheal intubation. A nomogram provided a list of doses of ketamine (in milligrams) that was based on patient weight, which corresponded to a full dose (2.0 mg per kilogram of body weight), an intermediate dose (1.5 mg per kilogram), or a reduced dose (1.0 mg per kilogram) (details are provided in the Supplementary Appendix).⁴⁹ Clinicians selected the dose of ketamine to administer.

For patients assigned to the etomidate group, clinicians were instructed to administer etomidate intravenously to induce anesthesia for tracheal intubation. A nomogram provided a list of doses of etomidate (in milligrams) that was based on patient weight, which corresponded to a full dose (0.3 mg per kilogram), an intermediate dose (0.25 mg per kilogram), or a reduced dose (0.2 mg per kilogram) (details are provided in the Supplementary Appendix). Clinicians selected the dose of etomidate to administer.

All other aspects of the patients' medical care were at the discretion of the treating clinicians, including the administration of vasopressors before induction, the choice of neuromuscular blocking agent for intubation, the approach to sedation and analgesia during mechanical ventilation, and the administration of intravenous fluids, vasopressors, and systemic glucocorticoids.

DATA COLLECTION

A trained observer who was not involved in the performance of the intubation collected data on the duration of intubation, the systolic blood pressure, and the administration of vasopressors during the interval between induction of anesthesia and 2 minutes after intubation. Immediately after

intubation, clinicians reported whether the patient had arrhythmia or cardiac arrest between the time of induction of anesthesia and 2 minutes after intubation. Trial personnel reviewed the medical record to collect data on the patients' baseline characteristics, periprocedural care, and clinical outcomes. They collected information on death from the medical record, from publicly available vital statistics records, and during follow-up telephone calls with patients or family members at 3 and 12 months (details are provided in the Supplementary Methods section of the Supplementary Appendix).

OUTCOMES

The primary outcome was in-hospital death by day 28, which was defined as death from any cause occurring between enrollment and 28 days after enrollment, with outcome ascertainment ending at hospital discharge. The single prespecified secondary outcome was cardiovascular collapse during intubation, which was defined by the occurrence of any of the following events during the interval between induction of anesthesia and 2 minutes after tracheal intubation: a systolic blood pressure below 65 mm Hg, receipt of a new or increased dose of vasopressors, or cardiac arrest. Exploratory procedural outcomes included the lowest systolic blood pressure, a lowest systolic blood pressure below 80 mm Hg, a highest systolic blood pressure above 180 mm Hg, the lowest oxygen saturation, and a lowest oxygen saturation below 80% — all of which were assessed during the interval between induction of anesthesia and 2 minutes after intubation — as well as successful intubation on the first attempt and the time from induction of anesthesia to intubation. Exploratory clinical outcomes were the number of ventilator-free days, vasopressor-free days, and ICU-free days. Safety outcomes included the systolic blood pressure at 24 hours after enrollment and ongoing receipt of vasopressors at 24 hours. A full list of the trial outcomes, including additional details, is provided in the Supplementary Appendix.

STATISTICAL ANALYSIS

Details regarding the determination of the sample size have been reported previously⁴⁸ and are included in the Supplementary Appendix. Assuming an incidence of in-hospital death by day 28 of 30.0% in the etomidate group,^{4,8} 80% statistical

power, and a two-sided alpha level of 0.05, we calculated that 2308 patients would need to be enrolled to detect an absolute difference of 5.2 percentage points between the groups in the incidence of in-hospital death by day 28. To ensure adequate power if data were missing in up to 3% of the patients, we planned to enroll a total of 2364 patients (1182 per group). The trial protocol included a single interim analysis, which was planned to be performed after 1182 patients had been enrolled; a P value threshold of 0.001 or less was to be used for the difference between the groups in the primary outcome that would justify stopping the trial. On the basis of the results of the interim analysis, the data and safety monitoring board recommended that the trial continue without modification.

The primary analysis was an intention-to-treat comparison of the primary outcome in the ketamine and etomidate groups that was performed with the use of a generalized linear mixed-effects model with a random effect for trial site and a fixed effect for trial-group assignment (ketamine group vs. etomidate group) without adjustment for covariates. The primary analysis included all the patients who had undergone randomization, except for those who had withdrawn from follow-up before ascertainment of the primary outcome. Sensitivity analyses of the primary outcome included the following: an unadjusted analysis that was performed with the use of a chi-square test; an analysis with adjustment for prespecified baseline covariates; analyses in which patients who had withdrawn before outcome ascertainment were classified as having died in the hospital by day 28 (the primary outcome) or as not having died in the hospital by day 28; an analysis of death by day 28 in any location (i.e., including both in-hospital deaths and deaths that occurred after hospital discharge); and an analysis of survival through day 28 in any location with the use of the Kaplan–Meier method (details are provided in the Supplementary Appendix).

In accordance with published guidelines,^{50,51} we examined whether prespecified baseline variables modified the effect of the trial-group assignment on the primary outcome with three approaches. The first approach involved subgroup analyses that used a generalized linear mixed-effects model with a random effect for trial site and fixed effects for trial group, the proposed effect modifier, and the interaction between the

effect modifier and the trial group, without adjustment for covariates. The second was a risk-modeling approach, in which previously validated models for the baseline risk of in-hospital death by day 28 were used.⁵² The third was an effect-modeling approach that used a machine-learning model to predict the effect of ketamine, as compared with etomidate, on the primary outcome for each patient on the basis of the patient's individual characteristics (individualized treatment effect).^{53,54} Details of these analyses are provided in the Supplementary Appendix.

Secondary and exploratory outcomes were compared in the two trial groups with the use of a chi-square test for categorical variables and the Wilcoxon rank-sum test for continuous or ordinal variables; between-group differences are reported as point estimates and 95% confidence intervals. The widths of the confidence intervals were not adjusted for multiplicity and should not be used to infer definitive differences in treatment effects between the two groups. All the analyses were performed with the use of R software, version 4.4.2 (R Foundation for Statistical Computing).

RESULTS

PATIENTS

Between April 6, 2022, and August 10, 2025, a total of 3439 patients were assessed for eligibility, of whom 2367 (68.8%) were enrolled and underwent randomization. The reasons for exclusion are listed in Figure S1 in the Supplementary Appendix. Two patients who were identified after enrollment as being prisoners were excluded from all analyses. The remaining 2365 patients were included in the trial population. The median age of the patients was 60 years, 46.7% had sepsis or septic shock, and 22.0% were receiving vasopressors (Table 1). Tracheal intubation was performed in an ED for 55.7% of the patients and in an ICU for 44.3% of the patients (Table S1). A total of 1176 patients were assigned to the ketamine group, and 1189 patients were assigned to the etomidate group (Tables S2 through S6). Six patients discontinued follow-up before day 28, and analyses of outcomes at day 28 included 1173 patients in the ketamine group and 1186 patients in the etomidate group. The representativeness of the trial population is described in the Supplementary Appendix.

MEDICATIONS FOR TRACHEAL INTUBATION

Of the 1176 patients in the ketamine group, 1167 (99.2%) received ketamine, and 1184 of the 1189 patients in the etomidate group (99.6%) received etomidate (Table 2 and Tables S7 and S8). The median dose of ketamine was 140 mg (interquartile range, 100 to 150), which was equivalent to 1.6 mg per kilogram (interquartile range, 1.4 to 2.0) (Fig. S2). The median dose of etomidate was 20 mg (interquartile range, 20 to 25), which was equivalent to 0.28 mg per kilogram (interquartile range, 0.24 to 0.31). Approximately 2% of the patients in each group received propofol, benzodiazepines, or opiates during the induction of anesthesia. A total of 99.7% of the patients in each of the two groups received a neuromuscular blocking agent (Table S9). Additional characteristics of the tracheal intubation procedure are shown in Table 2 and Tables S10 and S11.

PRIMARY OUTCOME

In-hospital death by day 28 occurred in 330 of 1173 patients (28.1%) in the ketamine group and in 345 of 1186 patients (29.1%) in the etomidate group (risk difference adjusted for trial site, -0.8 percentage points; 95% confidence interval [CI], -4.5 to 2.9; $P=0.65$) (Fig. 1 and Table 3). Death in any location by day 28 (one of the sensitivity analyses) occurred in 378 of 1173 patients (32.2%) in the ketamine group and in 384 of 1186 patients (32.4%) in the etomidate group (risk difference adjusted for trial site, 0.0 percentage points; 95% CI, -3.9 to 3.9). All the sensitivity analyses showed results similar to those of the main analysis (Fig. S3 and Tables S12 and S13), as did the subgroup analyses (Fig. 2 and Fig. S4), including the analysis of the subgroup comprising the 1101 patients with sepsis or septic shock (in-hospital death occurred in 38.8% of the patients in the ketamine group and in 38.2% of those in the etomidate group; risk difference adjusted for trial site, 1.0 percentage points; 95% CI, -4.8 to 6.7). Risk-modeling analyses and effect-modeling analyses did not show heterogeneity in the treatment effect (Figs. S5, S6, and S7 and Tables S14 and S15).

SECONDARY OUTCOME

Cardiovascular collapse during intubation occurred in 260 of 1176 patients (22.1%) in the ketamine group and in 202 of 1189 patients (17.0%) in the etomidate group (risk difference, 5.1 percentage

points; 95% CI, 1.9 to 8.3) (Table 3, Fig. S8, and Table S16). Among the patients with sepsis or septic shock, cardiovascular collapse occurred in 30.6% of the patients in the ketamine group and in 20.9% of those in the etomidate group (risk difference, 9.7 percentage points; 95% CI, 4.6 to 14.9). Among the patients with a high severity of illness (defined by an Acute Physiology and Chronic Health Evaluation [APACHE] II score of ≥ 20 ; scores range from 0 to 71, with higher scores indicating a greater severity of illness), cardiovascular collapse occurred in 31.4% of the patients in the ketamine group and in 20.7% of those in the etomidate group (risk difference, 10.7 percentage points; 95% CI, 5.5 to 16.0) (Fig. S9).

EXPLORATORY OUTCOMES

The median lowest systolic blood pressure during the interval between induction of anesthesia and 2 minutes after tracheal intubation was 112 mm Hg (interquartile range, 92 to 138) in the ketamine group and 118 mm Hg (interquartile range, 98 to 141) in the etomidate group (median difference, -6; 95% CI, -9 to -1) (Table 3 and Fig. S10). A total of 164 of 1138 patients (14.4%) in the ketamine group and 123 of 1156 patients (10.6%) in the etomidate group had a systolic blood pressure below 80 mm Hg during the interval between induction of anesthesia and 2 minutes after tracheal intubation (risk difference, 3.8 percentage points; 95% CI, 1.1 to 6.5). A decrease in systolic blood pressure of more than 30 mm Hg during this period occurred in 265 of 1109 patients (23.9%) in the ketamine group and in 165 of 1126 patients (14.7%) in the etomidate group (risk difference, 9.2 percentage points; 95% CI, 6.0 to 12.5).

The median time from induction of anesthesia to successful intubation was 112 seconds (interquartile range, 86 to 155) in the ketamine group, as compared with 103 seconds (interquartile range, 80 to 134) in the etomidate group (median difference, 9; 95% CI, 5 to 14). The median number of ventilator-free days, vasopressor-free days, and ICU-free days appeared to be similar in the two groups, as did the remaining exploratory outcomes (Table 3; Tables S17, S18, and S19; and Fig. S11).

SAFETY OUTCOMES

Ventricular tachycardia during the interval between induction of anesthesia and 2 minutes after intubation (a post hoc outcome added after the enroll-

Table 1. Characteristics of the Patients at Baseline.*

Characteristic	Ketamine (N = 1176)	Etomidate (N = 1189)
Median age (IQR) — yr	60 (45–69)	60 (44–69)
Female sex — no. (%)	498 (42.3)	492 (41.4)
Race or ethnic group — no. (%)†		
Non-Hispanic White	686 (58.3)	706 (59.4)
Non-Hispanic Black	300 (25.5)	287 (24.1)
Hispanic	130 (11.1)	132 (11.1)
Other	60 (5.1)	64 (5.4)
Median weight (IQR) — kg	78.9 (65.1–95.6)	78.5 (65.3–93.3)
Median body-mass index (IQR)‡	26.9 (23.0–32.4)	26.7 (22.5–32.1)
Location of intubation — no. (%)		
Emergency department	663 (56.4)	655 (55.1)
Intensive care unit	513 (43.6)	534 (44.9)
Chronic conditions — no. (%)		
Adrenal insufficiency or long-term receipt of glucocorticoids	136 (11.6)	129 (10.8)
Cirrhosis	165 (14.0)	166 (14.0)
Congestive heart failure	175 (14.9)	158 (13.3)
Coronary artery disease	141 (12.0)	153 (12.9)
Hypertension	536 (45.6)	533 (44.8)
Cancer	227 (19.3)	215 (18.1)
Acute conditions — no. (%)§		
Acute cardiac condition¶	216 (18.4)	223 (18.8)
Acute respiratory condition	678 (57.7)	683 (57.4)
Acute neurologic condition**	125 (10.6)	121 (10.2)
Sepsis or septic shock††	539 (45.8)	565 (47.5)
Median Glasgow Coma Scale score (IQR)‡‡	11 (7–15)	11 (7–15)
Median APACHE II score (IQR)§§	18 (13–24)	18 (13–24)
Median highest heart rate within 1 hr before enrollment (IQR) — beats per min	107 (90–125)	108 (92–126)
Median lowest systolic blood pressure within 1 hr before enrollment (IQR) — mm Hg¶¶	115 (96–136)	114 (94–135)
Receipt of vasopressors within 1 hr before enrollment — no. (%)	246 (20.9)	274 (23.0)

* Percentages may not total 100 because of rounding. IQR denotes interquartile range.

† Race and ethnic group were reported by the patients or their surrogates as part of clinical care and were obtained from the electronic health record by research personnel and grouped into fixed categories.

‡ Data on body-mass index (the weight in kilograms divided by the square of the height in meters) were missing for 19 patients (0.8%): 8 in the ketamine group and 11 in the etomidate group.

§ Data on acute conditions were abstracted from the electronic health record and grouped into prespecified categories. Patients could have had more than one acute condition.

¶ Acute cardiac condition was defined as the presence of one or more of the following conditions at the time of enrollment: cardiac arrest, cardiogenic shock, congestive heart failure, cardiogenic pulmonary edema, pulmonary hypertension, or myocardial infarction.

|| Acute respiratory condition was defined as the presence of one or more of the following conditions at the time of enrollment: acute respiratory distress syndrome, hypercapnic respiratory failure, hypoxemic respiratory failure, or pneumonia.

** Acute neurologic condition was defined as the presence of one or more of the following conditions at the time of enrollment: intracranial bleeding, meningitis, encephalitis, or stroke.

†† Sepsis or septic shock at enrollment was defined according to the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3).

‡‡ Glasgow Coma Scale scores range from 3 to 15, with lower scores indicating a lower level of consciousness. Glasgow Coma Scale scores were missing for 10 patients (0.4%): 6 in the ketamine group and 4 in the etomidate group.

§§ Scores on the Acute Physiology and Chronic Health Evaluation (APACHE) II range from 0 to 71, with higher scores indicating a greater severity of illness.

¶¶ Data on the lowest systolic blood pressure within 1 hour before enrollment were missing for 22 patients (0.9%): 11 in the ketamine group and 11 in the etomidate group.

Table 2. Characteristics of the Intubation Procedure.

Characteristic	Ketamine (N = 1176)	Etomidate (N = 1189)	Difference (95% CI)*
Primary medication for induction of anesthesia — no. (%)†			
Ketamine	1167 (99.2)	3 (0.3)	99.0 (98.4 to 99.6)
Etomidate	6 (0.5)	1184 (99.6)	−99.1 (−99.6 to −98.5)
None	3 (0.3)	2 (0.2)	0.1 (−0.3 to 0.5)
Neuromuscular blocking agent — no. (%)‡			
Rocuronium	810 (69.0)	819 (69.0)	0.0 (−3.7 to 3.7)
Succinylcholine	362 (30.8)	365 (30.7)	0.1 (−3.6 to 3.8)
None	3 (0.3)	3 (0.3)	0.0 (−0.4 to 0.4)
Measurements or treatments at induction of anesthesia			
Median oxygen saturation (IQR) — %§	99 (97–100)	99 (97–100)	0 (−1 to 1)
Preoxygenation — no. (%)	1172 (99.7)	1186 (99.7)	−0.1 (−0.5 to 0.4)
Median systolic blood pressure (IQR) — mm Hg¶	127 (110–147)	127 (110–148)	0 (−3 to 3)
Vasopressor bolus or increased infusion rate — no. (%)	207 (17.6)	234 (19.7)	−2.1 (−5.2 to 1.1)
Laryngoscope used on the first attempt — no. (%)			
Video	1124 (95.6)	1127 (94.8)	0.8 (−0.9 to 2.5)
Direct	49 (4.2)	60 (5.0)	−0.9 (−2.6 to 0.8)
Other	3 (0.3)	2 (0.2)	0.1 (−0.3 to 0.5)
Instrument used on the first intubation attempt — no. (%)**			
Endotracheal tube with stylet	672 (57.3)	689 (58.1)	−0.8 (−4.7 to 3.2)
Bougie	455 (38.8)	446 (37.6)	1.2 (−2.7 to 5.1)
Neither endotracheal tube with stylet nor bougie	45 (3.8)	51 (4.3)	−0.5 (−2.1 to 1.1)

* For categorical variables, the unadjusted difference in the percentage is reported in percentage points, and for continuous variables, the unadjusted difference in the median value is reported.

† One patient in the ketamine group received ketamine and then also received etomidate as a second induction medication. A total of 35 patients (1.5%) received either propofol or a benzodiazepine in addition to ketamine or etomidate: 18 in the ketamine group and 17 in the etomidate group. A total of 5 patients had hemodynamic instability or cardiac arrest before induction and underwent intubation without an induction medication: 3 in the ketamine group and 2 in the etomidate group.

‡ One patient in the ketamine group received both rocuronium and succinylcholine. Data on neuromuscular blocking agents were missing in 4 patients (0.2%): 2 in the ketamine group and 2 in the etomidate group.

§ Data on oxygen saturation at induction were missing in 108 patients (4.6%): 54 in the ketamine group and 54 in the etomidate group.

¶ Data on systolic blood pressure at induction were missing in 112 patients (4.7%): 59 in the ketamine group and 53 in the etomidate group.

|| Three patients in the ketamine group and 2 patients in the etomidate group underwent intubation with the use of a bronchoscope or an intubating laryngeal mask airway.

** Among the 2360 patients who underwent intubation with a laryngoscope, data on the instrument that was used on the first attempt were missing in 2 patients (0.1%): 1 in the ketamine group and 1 in the etomidate group.

ment of the first 567 patients) occurred in 9 of 884 patients (1.0%) in the ketamine group and in 2 of 905 patients (0.2%) in the etomidate group (risk difference, 0.8 percentage points; 95% CI, 0.1 to 1.5) (Table S20). A total of 420 of 1079 patients (38.9%) in the ketamine group and 458 of 1082 patients (42.3%) in the etomidate group were receiving vasopressors at 24 hours after enrollment (risk difference, −3.4 percentage points; 95% CI, −7.5 to 0.7).

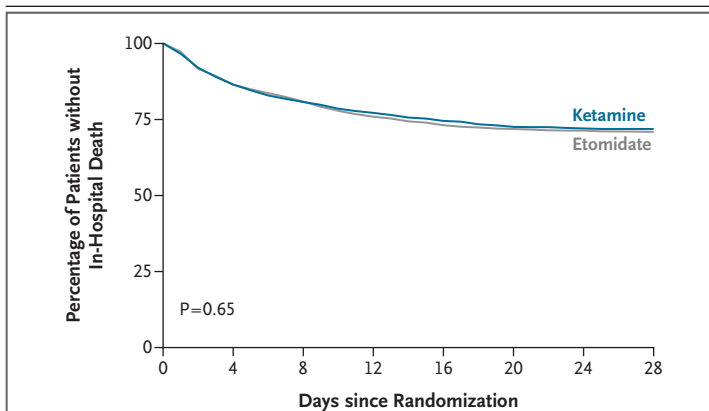


Figure 1. In-Hospital Death, According to Trial Group.

Shown is the percentage of patients without in-hospital death from enrollment through day 28. The incidence of in-hospital death by day 28 (the primary outcome) did not differ significantly between the ketamine group (28.1%) and the etomidate group (29.1%) (risk difference adjusted for trial site, -0.8 percentage points; 95% CI, -4.5 to 2.9 ; $P=0.65$ by a generalized linear mixed-effects model with a random effect for trial site).

DISCUSSION

Among critically ill adults undergoing emergency tracheal intubation in this multicenter, randomized trial, the use of ketamine for induction of anesthesia did not result in a significantly lower incidence of death than etomidate. The incidence of cardiovascular complications during intubation appeared to be higher with ketamine than with etomidate. These findings are important, given the concern that etomidate-induced corticosteroid insufficiency could affect a patient's risk of death — a concern that has led regulators to limit the availability of etomidate and has led clinicians to treat patients with induction medications that are more likely to lead to hemodynamic instability.

Numerous studies have shown that the use of etomidate to induce anesthesia impairs cortisol production for up to 72 hours.¹⁴⁻¹⁹ Whether this impairment in cortisol production increases the risk of death, however, has been unclear.^{55,56} In large observational studies, a lower risk of death was observed among critically ill adults who received ketamine than among those who received etomidate.^{7,17} Previous randomized trials have shown inconclusive results. Among 801 patients at a single center in a recent trial, the incidence of death by day 7 was 7.8 percentage points (95%

CI, 2.4 to 13.0) lower with ketamine than with etomidate, but the incidence of death by day 28 did not differ significantly between the groups.⁴¹ In a previous multicenter trial involving 469 patients, the incidence of death by day 28 did not differ significantly between the group that received ketamine and the group that received etomidate, but the incidence of organ dysfunction and death appeared to be lower in the ketamine group than in the etomidate group among patients with sepsis.¹⁸ Two recent meta-analyses that included data from all 2384 patients who had been enrolled in previous randomized trials reached differing conclusions, with one showing that the use of ketamine probably led to a lower risk of death than etomidate⁴³ and one showing no difference.⁴⁴

The concern stemming from these earlier studies that the use of etomidate might increase patients' risk of death has led to the removal of etomidate from the market in some countries²³⁻²⁵ and to recommendations against its use in some guidelines.²⁹ Among the 2365 patients in the current trial — approximately as many patients as were included in the aforementioned previous randomized trials combined — we observed no significant difference in the incidence of death between the ketamine and etomidate groups at any time point or in any subgroup.

One reason that ketamine has been increasingly used for induction of anesthesia in critically ill adults^{7,26,27} is the perception that ketamine maintains hemodynamic stability during intubation better than other induction medications.^{24,28,29} Some experts specifically recommend ketamine for intubation of patients with sepsis or those who are receiving vasopressors.^{24,29,57,58} Our trial showed that hypotension, receipt of vasopressors, and ventricular tachycardia during intubation each appeared to be more common with ketamine than with etomidate. These effects appeared to be largest among patients with sepsis and among those receiving vasopressors. Although discordant with the recommendations in some guidelines,²⁹ these findings are concordant with the results of two observational studies and one previous randomized trial, each of which showed a higher incidence of cardiovascular complications with ketamine than with etomidate.^{33,34,41}

Our trial has several strengths. The design included randomization to balance baseline characteristics, enrollment of a large sample of pa-

Table 3. Outcomes.

Outcome	Ketamine (N=1176)	Etomidate (N=1189)	Difference (95% CI)*
Primary outcome: in-hospital death from any cause by day 28 — no. (%)†	330 (28.1)	345 (29.1)	−0.8 (−4.5 to 2.9)‡
Secondary outcome: cardiovascular collapse during the interval between induction of anesthesia and 2 minutes after intubation — no. (%)	260 (22.1)	202 (17.0)	5.1 (1.9 to 8.3)
Systolic blood pressure <65 mm Hg§	73 (6.4)	64 (5.5)	0.9 (−1.0 to 2.8)
Receipt of a new or increased dose of vasopressors	251 (21.3)	189 (15.9)	5.4 (2.3 to 8.6)
Cardiac arrest¶	12 (1.0)	10 (0.8)	0.2 (−0.6 to 1.0)
Exploratory procedural outcomes			
Median lowest systolic blood pressure (IQR) — mm Hg§	112 (92–138)	118 (98–141)	−6 (−9 to −1)
Lowest systolic blood pressure <80 mm Hg — no. (%)§	164 (14.4)	123 (10.6)	3.8 (1.1 to 6.5)
Median highest systolic blood pressure (IQR) — mm Hg§	140 (115–164)	141 (118–168)	−2 (−7 to 2)
Highest systolic blood pressure >180 mm Hg — no. (%)§	154 (13.5)	191 (16.5)	−2.9 (−5.9 to 0.0)
Median lowest oxygen saturation (IQR) — %	97 (90–100)	97 (89–100)	0 (0 to 2)
Lowest oxygen saturation <80% — no. (%)	125 (11.1)	126 (11.1)	0.0 (−2.6 to 2.6)
Successful intubation on the first attempt — no. (%)**	1005 (85.7)	1029 (86.7)	−1.0 (−3.8 to 1.8)
Median time from induction of anesthesia to intubation (IQR) — sec††	112 (86–155)	103 (80–134)	9 (5 to 14)
Exploratory clinical outcomes‡‡			
Median ventilator-free days (IQR)	23 (0–26)	23 (0–26)	0 (−1 to 1)
Median vasopressor-free days (IQR)	25 (0–28)	25 (0–28)	0 (−1 to 1)
Median ICU-free days (IQR)	20 (0–24)	19 (0–24)	1 (−1 to 2)
Safety outcomes			
Median systolic blood pressure at 24 hours (IQR) — mm Hg§§	114 (102–130)	114 (103–129)	0 (−2 to 3)
Ongoing receipt of vasopressors at 24 hours — no. (%)¶¶	420 (38.9)	458 (42.3)	−3.4 (−7.5 to 0.7)

* For the primary outcome, the absolute risk difference with adjustment for trial site, which was generated with the use of a general linear mixed-effects model with a random effect for trial site without adjustment for covariates, is shown. For other outcomes, the unadjusted absolute risk difference in the percentage is reported in percentage points for categorical variables, and the unadjusted difference in the median value is reported for continuous and ordinal variables; the widths of the confidence intervals were not adjusted for multiplicity and should not be used to infer definitive differences in treatment effects between the two groups.

† A total of 6 patients (0.3%) withdrew from follow-up before day 28 and were missing data for 28-day outcomes: 3 in the ketamine group and 3 in the etomidate group.

‡ P=0.65.

§ Data on systolic blood pressure during the interval between induction of anesthesia and 2 minutes after intubation were missing for 68 patients (2.9%): 38 in the ketamine group and 30 in the etomidate group.

¶ Of the 22 patients (0.9%) who had cardiac arrest during the interval between induction of anesthesia and 2 minutes after intubation, 5 (0.2%) died within 1 hour after intubation: 3 in the ketamine group and 2 in the etomidate group.

|| Data on the oxygen saturation during the interval between induction of anesthesia and 2 minutes after intubation were missing for 106 patients (4.5%): 49 in the ketamine group and 57 in the etomidate group.

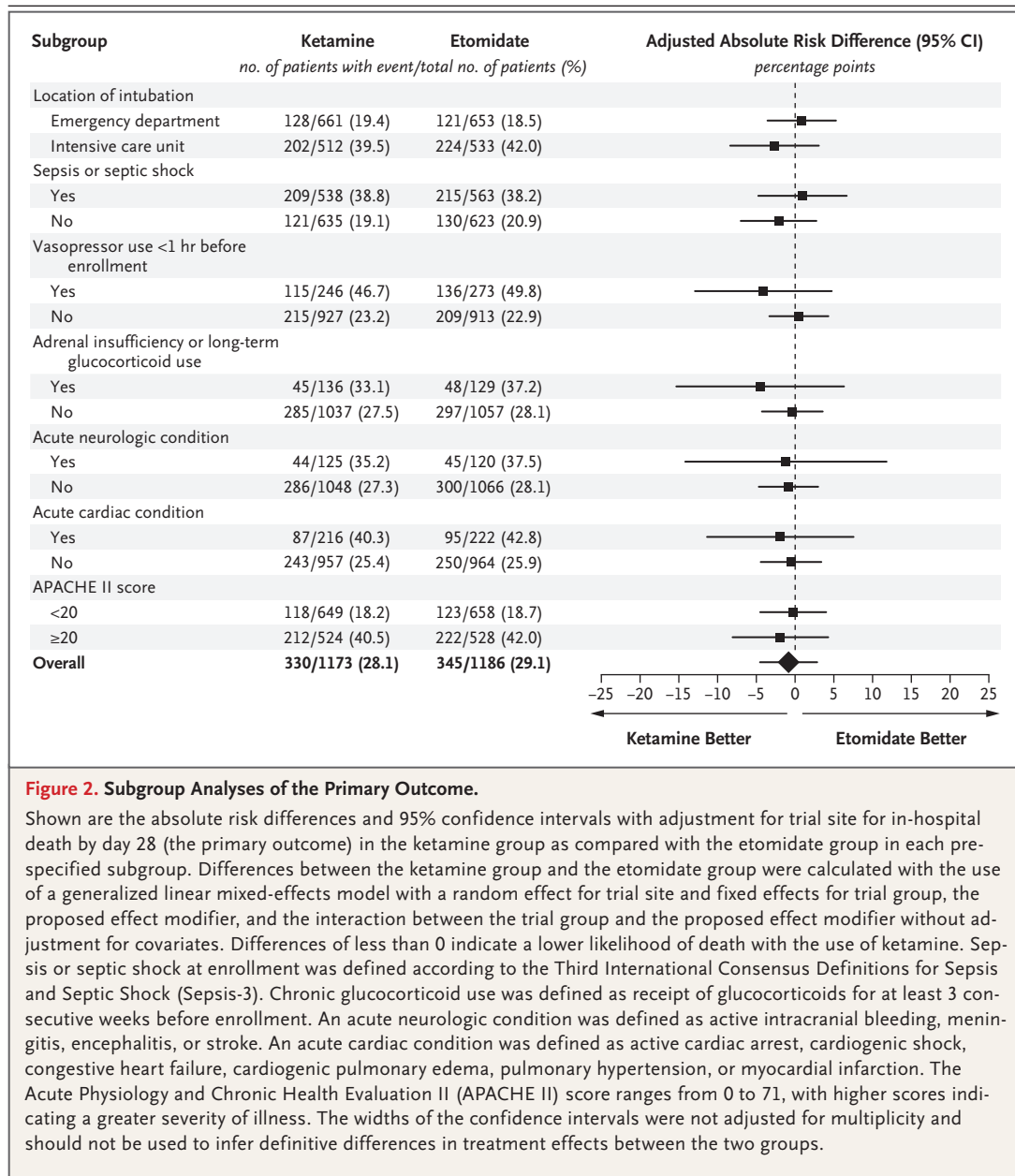
** This analysis includes the 2360 patients who underwent intubation with a laryngoscope.

†† Data on the time from induction of anesthesia to intubation were missing for 26 patients (1.1%): 14 in the ketamine group and 12 in the etomidate group.

‡‡ Ventilator-free days, vasopressor-free days, and ICU-free days were assessed at 28 days, with follow-up data censored at the time of hospital discharge.

§§ Data on systolic blood pressure at 24 hours after enrollment were unavailable for the 204 patients (8.6%) who died or were discharged before 24 hours. These data were missing for 9 patients (0.4%): 4 in the ketamine group and 5 in the etomidate group.

¶¶ Data on the ongoing receipt of vasopressors at 24 hours after enrollment were unavailable for the 204 patients (8.6%) who died or were discharged before 24 hours; 97 were in the ketamine group and 107 were in the etomidate group. Among the 878 patients who were receiving vasopressors at 24 hours after enrollment, the median vasopressor dose in norepinephrine equivalents (a post hoc outcome) was 0.09 µg per kilogram per minute (IQR, 0.04 to 0.20) in the ketamine group and 0.12 µg per kilogram per minute (IQR, 0.06 to 0.27) in the etomidate group (median difference, −0.03; 95% CI, −0.06 to −0.01).



tients to provide sufficient statistical power to detect clinically meaningful differences between the trial groups in the incidence of death, and conduct of the trial in EDs and ICUs at multiple centers to increase generalizability. The trial population had a sufficiently high severity of illness (nearly 30% of the patients died by day 28) and a sufficiently large number of patients with sepsis or septic shock (more than 1100) or who were receiving vasopressors (more than 500) to

credibly evaluate the hypothesized relationship between receipt of etomidate and death. Adherence to the trial-group assignments was excellent, and the percentage of patients with missing data for the primary outcome was low.

Our trial also has limitations. The trial excluded patients who presented with a primary diagnosis of trauma, so the results may not apply to such patients. Because the trial was not blinded, awareness of the trial-group assignments could have

influenced the trained observers' assessments of intraprocedural outcomes or clinicians' subsequent treatment decisions. Our findings do not rule out the possibility of small differences in outcomes in favor of either ketamine or etomidate, and for such commonly used medications, even small differences in outcomes might be clinically meaningful. Our trial compared ketamine and etomidate, so the results do not inform the effectiveness or safety of other induction medications, such as propofol, benzodiazepines, or barbiturates.

Among critically ill adults undergoing tracheal intubation, the use of ketamine to induce anesthesia did not lead to a significantly lower incidence of in-hospital death by day 28 than etomidate.

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REFERENCES

- Adhikari NKJ, Fowler RA, Bhagwanjee S, Rubenfeld GD. Critical care and the global burden of critical illness in adults. *Lancet* 2010;376:1339-46.
- Jivraj NK, Hill AD, Shieh M-S, et al. Use of mechanical ventilation across 3 countries. *JAMA Intern Med* 2023;183:824-31.
- Russotto V, Myatra SN, Laffey JG, et al. Intubation practices and adverse peri-intubation events in critically ill patients from 29 countries. *JAMA* 2021;325:1164-72.
- Gibbs KW, Semler MW, Driver BE, et al. Noninvasive ventilation for preoxygenation during emergency intubation. *N Engl J Med* 2024;390:2165-77.
- Brown CA III, Bair AE, Pallin DJ, Walls RM. Techniques, success, and adverse events of emergency department adult intubations. *Ann Emerg Med* 2015;65(4):363-370.e1.
- Kei J, Eurick T, Hauck TA. Intubation practices in community emergency departments. *Ann Emerg Med* 2025;86:169-74.
- Wunsch H, Bosch NA, Law AC, et al. Evaluation of etomidate use and association with mortality compared with ketamine among critically ill patients. *Am J Respir Crit Care Med* 2024;210:1243-51.
- Prekker ME, Driver BE, Trent SA, et al. Video versus direct laryngoscopy for tracheal intubation of critically ill adults. *N Engl J Med* 2023;389:418-29.
- Driver BE, Trent SA, Prekker ME, Reardon RF, Brown CA III. Sedative dose for rapid sequence intubation and post-intubation hypotension: is there an association? *Ann Emerg Med* 2023;82:417-24.
- Mohr NM, Santos Leon E, Carlson JN, et al. Endotracheal intubation strategy, success, and adverse events among emergency department patients during the COVID-19 pandemic. *Ann Emerg Med* 2023;81:145-57.
- Zane RD. Etomidate or midazolam for rapid sequence induction in patients with suspected sepsis? *NEJM Journal Watch*. December 17, 2010 (<https://www.jwatch.org/em201012170000001/2010/12/17/etomidate-or-midazolam-rapid-sequence-induction>).
- Williams LM, Boyd KL, Fitzgerald BM. Etomidate. In: StatPearls. Treasure Island, FL: StatPearls Publishing, 2025 (<http://www.ncbi.nlm.nih.gov/books/NBK535364/>).
- Budde AO, Mets B. Pro: etomidate is the ideal induction agent for a cardiac anesthetic. *J Cardiothorac Vasc Anesth* 2013;27:180-3.
- Wagner RL, White PF, Kan PB, Rosenthal MH, Feldman D. Inhibition of adrenal steroidogenesis by the anesthetic etomidate. *N Engl J Med* 1984;310:1415-21.
- Allolio B, Dörr H, Stuttmann R, Knorr D, Engelhardt D, Winkelmann W. Effect of a single bolus of etomidate upon eight major corticosteroid hormones and plasma ACTH. *Clin Endocrinol (Oxf)* 1985;22:281-6.
- Vinclair M, Broux C, Faure P, et al. Duration of adrenal inhibition following a single dose of etomidate in critically ill patients. *Intensive Care Med* 2008;34:714-9.
- Cuthbertson BH, Sprung CL, Annane D, et al. The effects of etomidate on adrenal responsiveness and mortality in patients with septic shock. *Intensive Care Med* 2009;35:1868-76.
- Jabre P, Combes X, Lapostolle F, et al. Etomidate versus ketamine for rapid sequence intubation in acutely ill patients: a multicentre randomised controlled trial. *Lancet* 2009;374:293-300.
- Lu Z, Zheng H, Chen Z, et al. Effect of etomidate vs propofol for total intravenous anesthesia on major postoperative complications in older patients: a randomized clinical trial. *JAMA Surg* 2022;157:888-95.
- Chan CM, Mitchell AL, Shorr AF. Etomidate is associated with mortality and adrenal insufficiency in sepsis: a meta-analysis. *Crit Care Med* 2012;40:2945-53.
- Hunter BR, Kirschner J. In patients with severe sepsis, does a single dose of etomidate to facilitate intubation increase mortality? *Ann Emerg Med* 2013;61:571-2.
- Dellinger RP, Levy MM, Rhodes A, et al. Surviving sepsis campaign: international guidelines for management of severe sepsis and septic shock: 2012. *Crit Care Med* 2013;41:580-637.
- Morris C, McAllister C. Etomidate for emergency anaesthesia; mad, bad and dangerous to know? *Anaesthesia* 2005;60:737-40.
- Morris C, Perris A, Klein J, Mahoney P. Anaesthesia in haemodynamically compromised emergency patients: does ketamine represent the best choice of induction agent? *Anaesthesia* 2009;64:532-9.
- DeMasi S, Self WH, Jerome RN, et al. Regulatory approval for the use of etomidate and ketamine for emergency tracheal intubation globally. *Acad Emerg Med* 2025;32:Suppl 1:68-9. abstract (<https://onlinelibrary.wiley.com/doi/10.1111/acem.70052?af=R>).
- Pollack MA, Fenati GM, Pennington TW, et al. The use of ketamine for air medical rapid sequence intubation was not associated with a decrease in hypotension or cardiopulmonary arrest. *Air Med J* 2020;39:111-5.
- Upchurch CP, Grijalva CG, Russ S, et al. Comparison of etomidate and ketamine for induction during rapid sequence intubation of adult trauma patients. *Ann Emerg Med* 2017;69(1):24-33.e2.
- Spotof H, Korshin JD, Sørensen MB, Skovsted P. The cardiovascular effects of ketamine used for induction of anaesthesia in patients with valvular heart disease. *Can Anaesth Soc J* 1979;26:463-7.
- Higgs A, McGrath BA, Goddard C, et al. Guidelines for the management of tracheal intubation in critically ill adults. *Br J Anaesth* 2018;120:323-52.
- Waxman K, Shoemaker WC, Lippmann M. Cardiovascular effects of anesthetic induction with ketamine. *Anesth Analg* 1980;59:355-8.
- Gelissen HPMM, Epema AH, Henning RH, Krijnen HJ, Hennis PJ, den Hertog A. Inotropic effects of propofol, thiopental, midazolam, etomidate, and ketamine on isolated human atrial muscle. *Anesthesiology* 1996;84:397-403.
- Christ G, Mundigler G, Merhaut C, et al. Adverse cardiovascular effects of ketamine infusion in patients with catecholamine-dependent heart failure. *Anaesth Intensive Care* 1997;25:255-9.
- April MD, Arana A, Schauer SG, et al. Ketamine versus etomidate and peri-intubation hypotension: a National Emergency Airway Registry study. *Acad Emerg Med* 2020;27:1106-15.
- Mohr NM, Pape SG, Runde D, Kaji AH, Walls RM, Brown CA III. Etomidate use is associated with less hypotension than ketamine for emergency department sepsis intubations: a NEAR cohort study. *Acad Emerg Med* 2020;27:1140-9.
- Maia IWA, von Hellmann R, Silva LOJ, et al. Ketamine vs etomidate in emergency department intubations: the search for a hemodynamically neutral agent. *Ann Emerg Med* 2024;84:S77-S78. abstract ([https://www.annemergmed.com/article/S0196-0644\(24\)00624-3/abstract](https://www.annemergmed.com/article/S0196-0644(24)00624-3/abstract)).
- Sanjamala HSR, Zafar H, Rehman TAZU, Nasir Y, Khosla J, Murray C. From calm to cardiac chaos: a case of ketamine induced brugada pattern. *J Am Coll Cardiol* 2025;85:Suppl:3675 ([https://www.jacc.org/doi/10.1016/S0735-1097\(25\)04159-2](https://www.jacc.org/doi/10.1016/S0735-1097(25)04159-2)).
- Stukus KS, Przybyłowicz RW, Backes CH Jr, Cohen DM. Ventricular tachycardia after ketamine sedation for fracture reduction. *Pediatr Emerg Care* 2014;30:730-2.
- Dewhirst E, Frazier WJ, Leder M, Fraser DD, Tobias JD. Cardiac arrest following ketamine administration for rapid sequence intubation. *J Intensive Care Med* 2013;28:375-9.
- Huebinger R, Habrat D, Fritz CL, Harrell AJ, Barrett WJ. Association of ketamine administration with intubation and cardiac arrest for prehospital patients with behavioral and substance-related complaints in the US. *Am J Emerg Med* 2025;90:250-1.
- Abdalla IG, Dafalla M. A case of cardiac arrest following ketamine administration. *Sudan J Paediatr* 2024;24:188-91.
- Matchett G, Gasanova I, Riccio CA, et al. Etomidate versus ketamine for emergency endotracheal intubation: a randomized clinical trial. *Intensive Care Med* 2022;48:78-91.

42. Kotani Y, Piersanti G, Maiucci G, et al. Etomidate as an induction agent for endotracheal intubation in critically ill patients: a meta-analysis of randomized trials. *J Crit Care* 2023;77:154317.
43. Koroki T, Kotani Y, Yaguchi T, et al. Ketamine versus etomidate as an induction agent for tracheal intubation in critically ill adults: a Bayesian meta-analysis. *Crit Care* 2024;28:48.
44. Greer A, Hewitt M, Khazaneh PT, et al. Ketamine versus etomidate for rapid sequence intubation: a systematic review and meta-analysis of randomized trials. *Crit Care Med* 2025;53(2):e374-e383.
45. Bandyopadhyay A, Haldar P, Sawhney C, Singh A. Efficacy of ketamine versus etomidate for rapid sequence intubation, among critically ill patients in terms of mortality and success rate: a systematic review and meta-analysis of randomized controlled trials. *Clin Exp Emerg Med* 2025 August 13 (Epub ahead of print).
46. Casey JD, Janz DR, Russell DW, et al. Bag-mask ventilation during tracheal intubation of critically ill adults. *N Engl J Med* 2019;380:811-21.
47. Exception from informed consent requirements for emergency research, 21 C.F.R. § 50.24 (1996) (<https://www.ecfr.gov/current/title-21/part-50/section-50.24>).
48. DeMasi SC, Imhoff B, Lewis AA, et al. Protocol and statistical analysis plan for a multicenter randomized trial of ketamine vs etomidate for emergency tracheal intubation. *CHEST Crit Care* 2025;3:100177 ([https://www.chestcc.org/article/S2949-7884\(25\)00050-4/fulltext](https://www.chestcc.org/article/S2949-7884(25)00050-4/fulltext)).
49. Food and Drug Administration. Ketalar (ketamine hydrochloride) injection (https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/016812s040lbl.pdf).
50. Schandelmaier S, Briel M, Varadhan R, et al. Development of the Instrument to Assess the Credibility of Effect Modification Analyses (ICEMAN) in randomized controlled trials and meta-analyses. *CMAJ* 2020;192:E901-E906.
51. Kent DM, Paulus JK, van Klaveren D, et al. The predictive approaches to treatment effect heterogeneity (PATH) statement. *Ann Intern Med* 2020;172:35-45.
52. Muhs A, Imhoff B, Seitz KP, et al. Risk factors for and prediction of 28-day in-hospital mortality in critically ill adults undergoing tracheal intubation. *Am J Respir Crit Care Med* 2025;211:A1502. abstract (<https://www.atsjournals.org/doi/10.1164/ajrccm.2025.211.Abstracts.A1502>).
53. Munroe ES, Spicer A, Castellvi-Font A, et al. Evidence-based personalised medicine in critical care: a framework for quantifying and applying individualised treatment effects in patients who are critically ill. *Lancet Respir Med* 2025;13:556-68.
54. Buell KG, Spicer AB, Casey JD, et al. Individualized treatment effects of oxygen targets in mechanically ventilated critically ill adults. *JAMA* 2024;331:1195-204.
55. Flynn G, Shehabi Y. Pro/con debate: is etomidate safe in hemodynamically unstable critically ill patients? *Crit Care* 2012;16:227.
56. Sklar MC, Wijeyesundera DN. An expiration date for etomidate? *Am J Respir Crit Care Med* 2024;210:1178-80.
57. Peksa GD, Gottlieb M. Ketamine should be the preferred agent for rapid sequence intubation. *Ann Emerg Med* 2021;78:722-3.
58. Ahmed A, Azim A. Difficult tracheal intubation in critically ill. *J Intensive Care* 2018;6:49.

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